NIMBUSEDGE TECHNOLOGIES PVT. LTD.

Routing Protocol Case Study – March 2024 Incident

Q1. RIP v2 — Chennai HQ Campus

a) Understanding of Case

In March 2024, the newly provisioned Delivery department subnet at Chennai HQ became unreachable. The NOC found that the subnet was **16 router hops** away from the core because a temporary VLAN gateway chain had been introduced for a client onboarding project.

The NOC Change Log shows that the temporary VLAN gateway chain and new Delivery subnet were introduced before the incident. Since RIP v2 supports a maximum of **15 hops**, the 16-hop subnet was treated as unreachable. The issue was not detected automatically because monitoring did not identify silent routing failures; the first indication was a client ticket.

b) Application of RIP v2

RIP uses **hop count** as its routing metric.

1. The Delivery subnet became **16 hops** from the HQ core.
2. RIP's maximum usable metric is **15**.
3. RIP assigns a metric of **16** to an unreachable destination.
4. Therefore, the Delivery subnet was considered unreachable.
5. The route was excluded from the routing table.

Thus:

16 hops → RIP metric 16 → unreachable → route omitted

The RCA confirms that RIP treats metric 16 as infinite/unreachable.

c) Analysis

The fault produced no automatic alert because RIP silently treats a 16-hop destination as unreachable rather than generating a separate error. NimbusEdge's monitoring mainly detected physical link-down events and did not monitor silent routing anomalies.

This shows a structural limitation of distance-vector routing: a routing problem can exist even when the physical network is operational, without producing a clear alarm.

d) Solution Design

The immediate solution is to redesign the temporary VLAN/gateway structure so that the Delivery subnet remains within RIP's hop limit.

For long-term improvement, NimbusEdge should migrate the Chennai campus toward OSPF. However, RIP was originally reasonable because Chennai HQ was a **small, stable campus network**, as shown in the topology diagram.

e) Recommendation

Problem: Delivery subnet became 16 hops away.

Technical cause: RIP metric reached 16 and the route was considered unreachable.

Recommendation: Redesign the subnet/gateway structure immediately and gradually migrate the campus to OSPF.

Q2. IGRP — Bengaluru–Pune–Hyderabad Regional Link

a) Understanding of Case

An ISP-side fault affected the **Bengaluru–Pune fiber link**, reducing bandwidth and increasing delay. Although a healthier alternate route existed through **Pune–Hyderabad–Bengaluru**, IGRP continued to prefer the degraded path, causing application latency of more than **600 ms**.
The route was finally manually adjusted at 18:10 IST so that traffic used the alternate path.

b) Application of IGRP Metric

IGRP primarily uses **bandwidth and delay** to calculate its composite metric.

For the degraded Bengaluru–Pune link:

Bandwidth = 2 Mbps = 2000 Kbps

Using the standard bandwidth component:

10^7 / 2000 = 5000

Therefore, the composite metric can be represented as:

IGRP Metric = 256 × [5000 + Delay/10]

Because the ISP fault also increased delay, the overall metric increased.

The alternate Pune–Hyderabad–Bengaluru path was healthier, so its effective metric should have become preferable after recalculation.

The reference files do not provide exact delay and bandwidth values for every link on the alternate path, so an exact numerical comparison cannot be calculated without making assumptions.

c) Analysis

This incident was mainly a **convergence/recalculation lag**, not a failure of the IGRP metric concept.

The degraded link's bandwidth decreased and delay increased, so IGRP should have increased its metric and eventually preferred the alternate path. However, route recalculation and propagation took too long, allowing traffic to remain on the degraded link.

d) Solution Design

NimbusEdge could tune IGRP, but replacing it with a faster-converging protocol is more suitable for the regional network.

The RCA recommends evaluating replacement of IGRP as part of the broader OSPF migration.

IGRP was originally preferable to RIP because it considers **bandwidth and delay**, making it more suitable for a regional network with variable link conditions. However, its slow response during this incident reduced that advantage.

e) Recommendation

Problem: Bengaluru–Pune link degraded.

Technical cause: Increased bandwidth/delay caused slow metric recalculation and convergence.

Recommendation: Migrate the regional segment from IGRP toward OSPF for faster and more predictable convergence.

Q3. OSPF — Core Backbone Migration

a) Understanding of Case

During the OSPF migration, the Hyderabad router was configured with an **incorrect Area ID**. The NOC Change Log identifies this error in CHG-0245.
Because of the mismatch, the Hyderabad router could not form an OSPF adjacency with the Chennai backbone router. This created a routing black hole for Hyderabad traffic even though the physical link remained up.

b) Application of OSPF Concepts

OSPF neighbors must have compatible configuration parameters, including:

- Area ID
- Hello interval
- Dead interval
- Authentication
- Subnet mask

In this incident, the **Area ID was incorrect**.

Therefore:

Incorrect Area ID → OSPF adjacency fails → no proper link-state exchange → Hyderabad routes unavailable → routing black hole

The RCA specifically identifies the Area ID mismatch as the cause.

c) Analysis

This failure was harder to detect than a physical link failure because the physical link remained **UP**. Monitoring therefore did not see a link-down event.

The RCA also identified process gaps such as the absence of a formal validation window and insufficient monitoring of silent routing anomalies.

A single transcription error therefore reached production without sufficient change validation.

d) Solution Design

Technical Fix

Correct the Hyderabad router's Area ID and verify:

1. OSPF neighbor adjacency
2. Routing table
3. End-to-end connectivity
4. Hello/Dead intervals
5. Authentication and subnet configuration

The incident report confirms that the Area ID was corrected and adjacency restored.

Process Safeguard

NimbusEdge should use:

- Peer review before OSPF changes
- Area ID verification checklist
- Post-change adjacency verification
- End-to-end ping/traceroute testing

The RCA specifically recommends peer-reviewed Area ID assignment during the OSPF migration.

Why Multi-Area OSPF?

OSPF provides **fast convergence, link-state routing, hierarchical scalability, and more accurate network knowledge** than the RIP/IGRP setup being replaced.

The planned design uses **Area 0 at Chennai and Areas 1–3 for regional sites**.

e) Before/After

Before:

Incorrect Area ID → adjacency failure → routing black hole.

After:

Correct Area ID → adjacency established → link-state exchange → routes restored.

Q4. BGP — ISP Peering and Failover

a) Understanding of Case

NimbusEdge used **ISP-A as the primary ISP** and **ISP-B as the secondary ISP**. When ISP-A suffered a fiber cut, traffic was expected to automatically fail over to ISP-B.

However, automatic failover did not occur. Client-facing traffic stalled and a senior network architect had to manually force traffic through ISP-B.

b) Application of BGP Attributes

NimbusEdge used **LOCAL_PREF** to prefer ISP-A during normal operation. AS-PATH prepending was also part of the routing policy.

The problem was that the LOCAL_PREF and AS-PATH policy had **not been validated in a live failover scenario**. As a result, route selection did not move automatically to ISP-B as expected.

Expected Logic

Normal:

ISP-A → higher preference → selected

ISP-B → lower preference → backup

ISP-A failure:

ISP-A route unavailable/withdrawn → ISP-B becomes best available path.

c) Analysis

BGP failover is particularly important because it controls connectivity between NimbusEdge and external autonomous systems.

The failure chain was:

ISP-A fiber cut → failover policy failed → ISP-B not automatically selected → outbound traffic stalled → manual intervention → service disruption → SLA breach

The Client SLA Notice records up to **9 hours 13 minutes of downtime** and confirms that the **99.9% monthly SLA was breached**.

The main risk was that the BGP failover policy had not been regularly tested.

d) Solution Design

NimbusEdge should maintain the following policy:

Condition	ISP-A	ISP-B
Normal operation	Higher LOCAL_PREF / Preferred	Lower LOCAL_PREF / Backup
ISP-A failure	Route withdrawn/unavailable	Automatically preferred

The configuration should be tested during a controlled maintenance window.

The RCA recommends a **quarterly BGP failover test** to validate LOCAL_PREF and AS-PATH behaviour.

BGP is the correct protocol for dual-ISP connectivity because it is designed for **routing between autonomous systems**, whereas RIP, IGRP, and OSPF are primarily used for internal routing.

e) Recommendation

Problem: ISP-A failed but automatic failover did not occur.

Technical cause: Unvalidated BGP preference/failover policy.

Recommendation: Correct and validate LOCAL_PREF/AS-PATH policies and perform quarterly failover testing.

Overall Conclusion

The March 18, 2024 outage involved four different routing problems:

Protocol	**Failure**
RIP v2	Delivery subnet exceeded 15-hop limit
IGRP	Slow reconvergence after link degradation
OSPF	Incorrect Area ID prevented adjacency
BGP	Unvalidated failover policy prevented automatic ISP switching

The incident affected six enterprise clients and resulted in an SLA breach.

NimbusEdge should therefore:

1. Redesign the Chennai Delivery routing path or migrate from RIP.
2. Replace IGRP with a faster-converging protocol.
3. Complete the multi-area OSPF migration with peer-reviewed configuration.
4. Test BGP failover regularly.
5. Improve monitoring for silent routing failures and adjacency problems.

This combination addresses both the **technical routing failures and the change-management/monitoring gaps** identified in the RCA.